KATHY PILLAI KUMAR

Data, AI and Software Engineer | Data Engineering, Machine Learning, AI Agents, Cloud, Security
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PROFESSIONAL SUMMARY

Data, AI and software engineer with three years at IBM India Software Labs on watsonx.data, an enterprise lakehouse used by financial services and payments clients, spanning vector database (Milvus) integration, LLM and agentic AI features, Go REST APIs, data pipelines, query engine performance and security remediation. Builds and ships live products end to end: Voltwise (forecasting, 44.7% lower error), VKinder (live AI platform), Nexora and MuleWatch. Data Science Intern at ZynAI (healthcare ML), Monash Faculty of IT tutor and Student Ambassador, and trainer with Junior Engineers and Deadly Coders. Completing a Master of Data Science at Monash University in November 2026.

TECHNICAL SKILLS

Languages: Python, SQL, Go, Java, C++, JavaScript, Node.js, React.

Data engineering: Databricks, PySpark, Spark, Presto, Apache Iceberg, Delta Lake, Delta Live Tables, Unity Catalog, dbt, Airflow, Kafka, Pub/Sub, Dataflow, Azure Data Factory, ETL/ELT, data modelling, data quality.

Databases & warehouses: PostgreSQL, PostGIS, pgvector, Milvus, Snowflake, Redshift, Netezza, BigQuery, DuckDB, MySQL, Hive, Redis.

AI & machine learning: scikit-learn, LightGBM, Random Forest, YOLOv8, ONNX, SHAP, forecasting, RAG, LLMs, fine-tuning, AI agents, MCP, Ollama, LLM evaluation, Azure ML, MLflow.

Cloud, DevOps & security: AWS, Azure, GCP, IBM Cloud, Kubernetes, Docker, Terraform, AWS SAM, GitHub Actions, Maven, Git, OpenTelemetry, Loki, Grafana, Twistlock, Aqua, OSS and static scanning, patching pipelines.

PROFESSIONAL EXPERIENCE

Software Developer, watsonx.data | IBM India Software Labs | Kochi, India January 2022 to January 2025

- Developed backend components in Python and C++ to bring Milvus vector database capability into IBM watsonx.data, moving from a standalone external vector store to the platform-native Milvus service: connection setup, credential and SSL handling, test-connection validation and metadata discovery across SaaS, Cloud Pak for Data and standalone deployments.
- Built data transformation preparing enterprise documents and table metadata for vector search (chunking, embedding generation, schema mapping); built semantic-layer APIs that trigger enrichment runs and store results; generated natural-language descriptions of data assets with LLMs; contributed agentic AI tool calling with retrieval over vector search.
- Developed and maintained Go RESTful APIs for the platform REST server, including the Redis connector and Bring Your Own JDBC Driver (BYOJ) APIs with DELETE support; validated Test Connection across 8 connectors with Postman and end-to-end tests; built Python tooling for the LH-CLI.
- Developed and maintained ETL/ELT pipelines with Databricks, PySpark, Python and SQL within an enterprise platform processing approximately 100 GB to 1 TB of data daily for financial services and payments clients; built Delta Lake MERGE INTO and Delta Live Tables workflows with Unity Catalog governance, and integrated data with Snowflake, Amazon Redshift, IBM Netezza and Azure.
- Fixed Presto query engine bugs in Java (visual explain for CREATE queries, multi-dot identifiers) and an access-control bug in schema filtering; analysed JVM heap, GC and Alluxio caching across JDK versions; improved Presto query performance by 20% through caching enhancements contributed to open source; ran a feasibility study porting the Trino Iceberg connector.
- Security Analyst for six months: closed 42 findings (6 critical, 14 high, 22 medium) from OSS, static, Twistlock and Aqua scans; patched Presto and third-party Java libraries and container images through patching pipelines across three remediation cycles; security focal for the SaaS and Cloud Pak for Data teams; security QA for credentials and sensitive-data exposure; audit logging.
- Built an AWS lakehouse proof of concept (CSV in S3, Parquet via Glue crawler, Iceberg via CTAS, queried from Presto); configured OpenTelemetry with Loki and Grafana; contributed to QHMM diagnostics; wrote a Presto troubleshooting runbook and setup guides.
- Automated QA and data validation with Python scripts and CLI tools, cutting manual effort by 50% and QA processing time by 25%; tested Iceberg, Hive, MySQL and PostgreSQL connectors on-prem, SaaS and standalone; gathered feature and security requirements directly with client representatives at major Indian banks.

Data Science Intern | ZynAI | Sydney, remote August 2026 to November 2026

- Built healthcare machine learning workflows for MetaGamma (preprocessing, training, evaluation), integrated Google APIs for MetaPharm data ingestion and built dashboards on healthcare data.

Coding Instructor and Trainer | Junior Engineers (in partnership with Deadly Coders) | Melbourne August 2025 to present

- Deliver introductory data science, AI and machine learning sessions to First Nations (Aboriginal and Torres Strait Islander) learners and students from culturally diverse backgrounds; teach three weekly coding classes; trained four new instructors.

Student Ambassador and Programming Tutor | Faculty of IT, Monash University | Melbourne 2025 to present

- Programming tutor and Python bootcamp tutor (2025); Student Ambassador (March 2026 to present) supporting enquiries, admissions, orientation and events with Moodle and Blackboard.

Data and Insight Officer | Monash Data and AI Society | Melbourne 2025

- Ran stakeholder interviews and workshops, built business cases and led platform and vendor evaluations, including Medallion Architecture and PySpark workshops.

Customer Service Representative | Woolworths | Melbourne September 2025 to October 2026

- Resolved customer enquiries, handled transactions and supported daily reporting in a busy store.

SELECTED PROJECTS

Voltwise - battery capacity forecasting platform | Python, scikit-learn, dbt, DuckDB, FastAPI, Azure, GCP | Live demo | GitHub 2026

- Random Forest forecasting 10 cycles ahead that reduced MAE by 44.7% versus persistence across 560 held-out forecasts from four NASA cells (leave-one-battery-out validation); dbt Core pipelines, DuckDB warehouse and semantic metrics layer with 13 data-quality checks.
- Azure Data Factory and Data Lake Storage ingestion; BigQuery, Pub/Sub and Dataflow for real-time telemetry; Azure Machine Learning and MLflow; FastAPI, Pydantic and Ollama/Qwen2.5 tool-calling AI; automated tests, GitHub Actions and GitHub Pages.

VKinder - live data and AI platform for families | React, Python, SQL, Aurora PostgreSQL, AWS | Live site 2026

- Designed, built and deployed end to end a live platform helping Melbourne parents and child-health practitioners compare neighbourhoods on child development, health services and environment, with 2031 forecasts and a scenario simulator.
- AI layer: RAG score explainer, plain-English search, pgvector similarity search, an AI agent that diagnoses failed refreshes, an MCP server and a fine-tuned small language model; MLflow registry, drift monitoring, retraining, graph neural networks and spatial regression.

Nexora (DataMedic) - data incident investigation workspace | Live demo | GitHub 2026

- Interactive workspace that investigates data pipeline incidents, diagnoses root causes and plays back verified recovery steps on reproducible scenarios.

MuleWatch - payments fraud data platform | Kafka, Apache Iceberg, dbt, Airflow, LightGBM, FastAPI 2026

- Kafka to Apache Iceberg pipeline with data contracts and quarantine, 28 tested dbt models, LightGBM with graph features (recall 0.97 vs 0.45 for SQL rules at equal alert volume, PR-AUC 0.94, synthetic data), SHAP, FastAPI and Streamlit, an LLM case-note writer with guardrails, Airflow, OpenTelemetry and GitHub Actions CI.

CloudEcho - cloud-native computer vision service | YOLOv8, ONNX, FastAPI, Kubernetes, Terraform, GCP, Azure

- Wildlife-detection model served as a FastAPI service on Kubernetes with Terraform, Databricks Auto Loader ingestion, health checks, autoscaling and Locust load testing.

Sensory-Aware Router - serverless geospatial app | AWS Lambda, Aurora Serverless, PostGIS, Node.js, AWS SAM

- Scores routes on crowd, noise and lighting data with spatial SQL, deployed as infrastructure as code.

Driver Drowsiness Detection | Python, PySpark, computer vision 2022

- Final-year project using PySpark and Python for data processing to train a drowsiness detection model.

EDUCATION

Master of Data Science | Monash University February 2025 to November 2026

B.Tech (Honours), Computer Science | TKM College of Engineering, Kollam | CGPA: 8.61/10 July 2018 to May 2022

COMMUNICATION AND LEADERSHIP

- Coding Instructor and Trainer, Junior Engineers | August 2025 to present: in partnership with Deadly Coders, delivered introductory data science, AI and machine learning sessions to First Nations (Aboriginal and Torres Strait Islander) learners and students from culturally diverse backgrounds; trained four new instructors.
- Student Ambassador, Monash Faculty of IT | March 2026 to present.

AWARDS AND COMMUNITY

- Winner, Make-a-thon Hackathon, Cochin University (2019); Winner, IEDC Idea Pitching Contest, Kerala Startup Mission (2021); participant, UNIHACK 2025.
- Data and Insight Officer, Monash Data Science Club; Events Officer, Monash Coding Assistive Tech Team (2025); Programming Tutor, Monash Faculty of IT (2025).

PROFESSIONAL DEVELOPMENT

IBM Developer Jumpstart Practitioner (2024); Cybersecurity, Data Privacy and AI, IBM (2024); Security and Privacy by Design (2023); Enterprise Design Thinking - Essentials for AI (2022); Agile Explorer (2022).

Temporary Graduate visa (subclass 485) with full working rights from 15 November 2026. Current Victorian Working with Children Check. References available upon request.